# SG No. 180 $D_6^4$ $P6_222$ [ hexagonal ]

* plus set: $+[0,\ 0,\ 0]$

Table 1: Wyckoff site: 3a, site symmetry: 222

| No. | position | mapping |
|---|---|---|
| 1 | $[0,\ 0,\ 0]$ | [1,4,8,11] |
| 2 | $\left[0,\ 0,\ \frac{2}{3}\right]$ | [2,5,7,10] |
| 3 | $\left[0,\ 0,\ \frac{1}{3}\right]$ | [3,6,9,12] |

Table 2: Wyckoff site: 3b, site symmetry: 222

| No. | position | mapping |
|---|---|---|
| 1 | $\left[0,\ 0,\ \frac{1}{2}\right]$ | [1,4,8,11] |
| 2 | $\left[0,\ 0,\ \frac{1}{6}\right]$ | [2,5,7,10] |
| 3 | $\left[0,\ 0,\ \frac{5}{6}\right]$ | [3,6,9,12] |

Table 3: Wyckoff site: 3c, site symmetry: 222

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{2},\ 0,\ 0\right]$ | [1,4,8,11] |
| 2 | $\left[0,\ \frac{1}{2},\ \frac{2}{3}\right]$ | [2,5,7,10] |
| 3 | $\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{3}\right]$ | [3,6,9,12] |

Table 4: Wyckoff site: 3d, site symmetry: 222

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{2},\ 0,\ \frac{1}{2}\right]$ | [1,4,8,11] |
| 2 | $\left[0,\ \frac{1}{2},\ \frac{1}{6}\right]$ | [2,5,7,10] |
| 3 | $\left[\frac{1}{2},\ \frac{1}{2},\ \frac{5}{6}\right]$ | [3,6,9,12] |

Table 5: Wyckoff site: 6e, site symmetry: 2..

| No. | position | mapping |
|---|---|---|
| 1 | $[0,\ 0,\ z]$ | [1,4] |
| 2 | $\left[0,\ 0,\ z+\frac{2}{3}\right]$ | [2,5] |
| 3 | $\left[0,\ 0,\ z+\frac{1}{3}\right]$ | [3,6] |
| 4 | $\left[0,\ 0,\ \frac{2}{3}-z\right]$ | [7,10] |
| 5 | $[0,\ 0,\ -z]$ | [8,11] |
| 6 | $\left[0,\ 0,\ \frac{1}{3}-z\right]$ | [9,12] |

Table 6: Wyckoff site: `6f`, site symmetry: `2..`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{2},\ 0,\ z\right]$ | `[1,4]` |
| 2 | $\left[0,\ \frac{1}{2},\ z+\frac{2}{3}\right]$ | `[2,5]` |
| 3 | $\left[\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{3}\right]$ | `[3,6]` |
| 4 | $\left[0,\ \frac{1}{2},\ \frac{2}{3}-z\right]$ | `[7,10]` |
| 5 | $\left[\frac{1}{2},\ 0,\ -z\right]$ | `[8,11]` |
| 6 | $\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{3}-z\right]$ | `[9,12]` |

Table 7: Wyckoff site: `6g`, site symmetry: `.2.`

| No. | position | mapping |
|---|---|---|
| 1 | $[x,\ 0,\ 0]$ | `[1,8]` |
| 2 | $\left[0,\ x,\ \frac{2}{3}\right]$ | `[2,7]` |
| 3 | $\left[-x,\ -x,\ \frac{1}{3}\right]$ | `[3,9]` |
| 4 | $[-x,\ 0,\ 0]$ | `[4,11]` |
| 5 | $\left[0,\ -x,\ \frac{2}{3}\right]$ | `[5,10]` |
| 6 | $\left[x,\ x,\ \frac{1}{3}\right]$ | `[6,12]` |

Table 8: Wyckoff site: `6h`, site symmetry: `.2.`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[x,\ 0,\ \frac{1}{2}\right]$ | `[1,8]` |
| 2 | $\left[0,\ x,\ \frac{1}{6}\right]$ | `[2,7]` |
| 3 | $\left[-x,\ -x,\ \frac{5}{6}\right]$ | `[3,9]` |
| 4 | $\left[-x,\ 0,\ \frac{1}{2}\right]$ | `[4,11]` |
| 5 | $\left[0,\ -x,\ \frac{1}{6}\right]$ | `[5,10]` |
| 6 | $\left[x,\ x,\ \frac{5}{6}\right]$ | `[6,12]` |

Table 9: Wyckoff site: `6i`, site symmetry: `..2`

| No. | position | mapping |
|---|---|---|
| 1 | $[x,\ 2x,\ 0]$ | `[1,11]` |
| 2 | $\left[-2x,\ -x,\ \frac{2}{3}\right]$ | `[2,10]` |
| 3 | $\left[x,\ -x,\ \frac{1}{3}\right]$ | `[3,12]` |
| 4 | $[-x,\ -2x,\ 0]$ | `[4,8]` |
| 5 | $\left[2x,\ x,\ \frac{2}{3}\right]$ | `[5,7]` |
| 6 | $\left[-x,\ x,\ \frac{1}{3}\right]$ | `[6,9]` |

Table 10: Wyckoff site: `6j`, site symmetry: `..2`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[x,\ 2x,\ \frac{1}{2}\right]$ | `[1,11]` |
| 2 | $\left[-2x,\ -x,\ \frac{1}{6}\right]$ | `[2,10]` |

*continued ...*

Table 10

| No. | position | mapping |
|---|---|---|
| 3 | $\left[x,\, -x,\, \frac{5}{6}\right]$ | [3,12] |
| 4 | $\left[-x,\, -2x,\, \frac{1}{2}\right]$ | [4,8] |
| 5 | $\left[2x,\, x,\, \frac{1}{6}\right]$ | [5,7] |
| 6 | $\left[-x,\, x,\, \frac{5}{6}\right]$ | [6,9] |

Table 11: Wyckoff site: 12k, site symmetry: 1

| No. | position | mapping |
|---|---|---|
| 1 | $[x,\, y,\, z]$ | [1] |
| 2 | $\left[-y,\, x-y,\, z+\frac{2}{3}\right]$ | [2] |
| 3 | $\left[-x+y,\, -x,\, z+\frac{1}{3}\right]$ | [3] |
| 4 | $[-x,\, -y,\, z]$ | [4] |
| 5 | $\left[y,\, -x+y,\, z+\frac{2}{3}\right]$ | [5] |
| 6 | $\left[x-y,\, x,\, z+\frac{1}{3}\right]$ | [6] |
| 7 | $\left[y,\, x,\, \frac{2}{3}-z\right]$ | [7] |
| 8 | $[x-y,\, -y,\, -z]$ | [8] |
| 9 | $\left[-x,\, -x+y,\, \frac{1}{3}-z\right]$ | [9] |
| 10 | $\left[-y,\, -x,\, \frac{2}{3}-z\right]$ | [10] |
| 11 | $[-x+y,\, y,\, -z]$ | [11] |
| 12 | $\left[x,\, x-y,\, \frac{1}{3}-z\right]$ | [12] |